Comments for your project

Prof. Draguljić

March 7, 2026

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See the RMarkdown file for how to easily create Table of contents, list of figures, and list of tables.

What follows is a list of common mistakes that I compiled from different courses. Please address the mistakes in your work to make it more professional.

Stop by my office if ask a question beofre you turn in substandard work.

Setting global R options

To set global options like figure size, printing of R code, etc. use this R command.

```
knitr::opts_chunk$set(fig.width = 10, fig.height = 5, echo = TRUE, eval = TRUE)
```

Suspending warning messages

If you load the tidyverse package you get all sorts of warnings. They are not necesarily bad. It's just R telling what it's doing.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to b
```

These comments should not be in your final report. To supress them use *message = FALSE, warning = FALSE* options in the R chunk where you load the package.

```
library(tidyverse)
```

General comments

Read your report

Read your final report before you submit it! And read it in your pdf file and not your rmd file. I will be reading your pdf file, so if something looks good in rmd file but not in pdf file, then it does not look good.

Read. Your. Final. Submission!

More than once.

Provide detailed answers

For any question (where applicable) provide written and visual answers (figures, tables, etc.) The written and visual answers should be stand-alone. That is, after reading your written response I should be able to imagine what that scatterplot or boxplot looks like.

Creating figures

Figure 1 shows a bad looking plot. The same plot that looks decent is shown in Figure 2. Note how I referenced the figures here using LaTeX. Not how I also added figure labels that also show up in the list of figures.

```
plot(iris$Sepal.Length, iris$Petal.Length, cex.lab = 0.75, cex.axis = 0.75,
     main = "Petal vs. Sepal length")
```

```
plot(iris$Sepal.Length, iris$Petal.Length,
     xlim = c(4, 8), las = 1, ylim = c(1, 7),
     xlab = "Sepal length (cm)", ylab = "Petal length (cm)",
     cex.lab = 1.1, cex.axis = 1.1)
```

Here are some comments on the bad plot.

- Axis and tickmark labels are too small.
- Both axis ranges are too small, i.e. they do not cover all the data.
- Since both variables are on the same scale, the plot should be more square than rectangular.

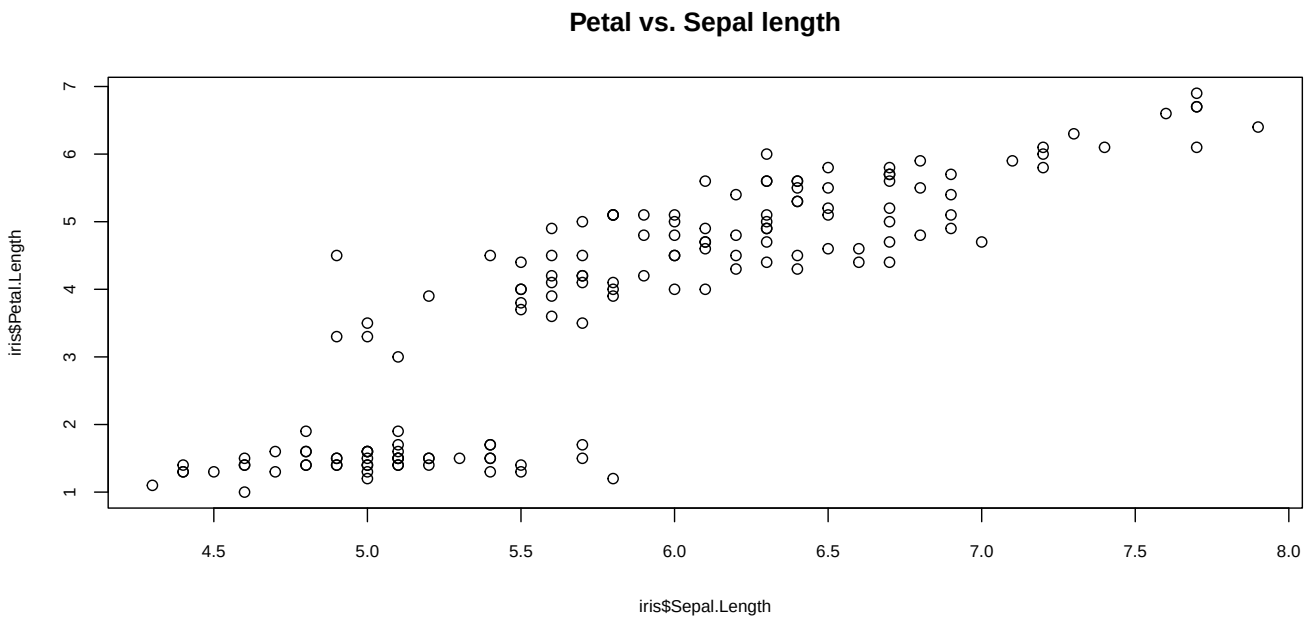

Figure 1: The scatterplot of Sepal.Length and Petla.lenghts

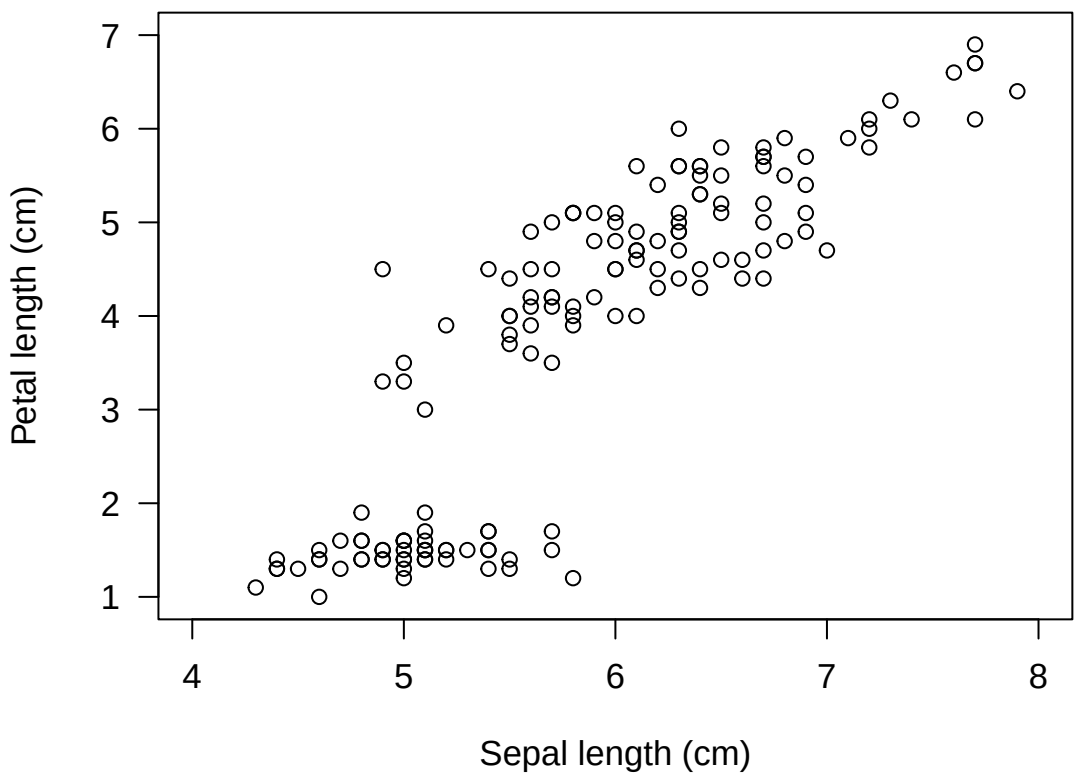

Figure 2: The relationship between sepal and petal lenghts

- Don't put a main title in your plot. That's what the figure label is for.
- Figure label
  - There are typos in there.
  - Never user R's variable names in your labels or, for that matter, in the text. Never! These are random names and might have no meaning relevant at all.
  - I see it's a scatterplot. No need to tell me that.
  - Don't capitalize "random" words. Start with a capital letter and capitalize what you would capitalize otherwise.
- Tickmark labels aren't rotated properly on the y-axis.
- Units are missing.

Keep your figures, generally, in the same format, i.e. the same label size, same tickmark size, etc., in the whole document.

Don't use colors that cannot be clearly seen in your plots. In particular, I see a lot of yellow plot symbols. They just tended to blur together on the page. So, yellow makes a particularly bad choice.

You don't need the legend if there is only one symbol or line on the plot. Two or more, add a legend.

Creating tables

- Add table labels. Don't make your labels too technical.

```
summaryStatIris <- iris %>%
  group_by(Species) %>%
  summarise(n = n(),
            avg = round(mean(Sepal.Length), 2),
            sd = round(sd(Sepal.Length), 2),
            lcb = round(avg - qt(0.975, n - 1)*sd/sqrt(n), 2),
            ucb = round(avg + qt(0.975, n - 1)*sd/sqrt(n), 2))
```

This is a bad table:

```
summaryStatIris
```

```
## # A tibble: 3 x 6
##   Species        n   avg    sd   lcb   ucb
##   <fct>      <int> <dbl> <dbl> <dbl> <dbl>
## 1 setosa        50  5.01  0.35  4.91  5.11
## 2 versicolor    50  5.94  0.52  5.79  6.09
## 3 virginica     50  6.59  0.64  6.41  6.77
```

Table 1 shows the summary statistics of the sepal length for each species. This table looks good. It is properly formatted and it has a table label that shows up in list of tables at the beginning.

```r
library(knitr)
kable(summaryStatIris,
      col.names = c("Species", "n", "Average", "Std. deviation",
                    "95% CI lower bound", "95% CI upper bound"),
      caption = "\\label{tbl:sumStats} The summary statistics for the sepal length by specie
```

Table 1: The summary statistics for the sepal length by species

Species	n	Average	Std. deviation	95% CI lower bound	95% CI upper bound
setosa	50	5.01	0.35	4.91	5.11
versicolor	50	5.94	0.52	5.79	6.09
virginica	50	6.59	0.64	6.41	6.77

Hard-coding the values

Do not hard-code values in your code. Eventually, this will be a bad news!

For example

```r
mean(iris$Petal.Length)
```

```
## [1] 3.758
```

```r
mean(iris$Sepal.Length)
```

```
## [1] 5.843333
```

```r
if(3.758 < 5.843333){
  junkVar <- iris$Petal.Length - (5.843333 - 3.758)
}
mean(junkVar)
```

```
## [1] 1.672667
```

should be coded as

```r
avgP <- mean(iris$Petal.Length)
avgS <- mean(iris$Sepal.Length)
if(avgP < avgS){
  junkVar <- iris$Petal.Length - (avgS - avgP)
}
mean(junkVar)
```

```
## [1] 1.672667
```

With the proper coding, if your data set changes, your code will still work. If you hard-code in the values, then you will have to go around and fix all those. You will miss some for sure.

Wrapping your R code

Keep your code on the page and in the shaded box.

```r
plot(iris$Sepal.Length, iris$Petal.Length, xlim = c(4, 8), las = 1, ylim = c(1, 7), xlab =
```

Nobody can read the code above. See Figure 2 code for proper and readable formatting.

Writing up the results

When you are describing a distribution, describe this distribution in terms of center or typical value (mean, median, or some other appropriate statistic), spread (usually standard deviation, IQR, or range), outliers, skewness, modality, etc.

When you're asked to compare two distributions, most often it makes sense to compare the average values. Saying something like

"on average, our data indicates that early class is getting higher scores than the afternoon class"

is much better than saying

"there is some overlap between boxplots for group A and group B."

Write in context of the problem

All your written explanations, interpretation, etc. should be in the context of a problem! They should also try to be as non-technical as possible.

While saying something like

"the distribution of the scores is skewed right"

is correct, saying something like

"majority of the countries have low GBGR scores indicating low acceptance of LGBTQ community"

will be much more appreciated by your non-statistical colleague.

Using R in the text

You can use R within your text. For example, suppose you want to say what the average sepal length in your data set is. You could do this by calculating the height first via

```
mean(iris$Sepal.Length)
```

```
## [1] 5.843333
```

and then, after copying and pasting the average height, saying something along the line of

"The average child's height is 5.84 cm."

Another option is to directly calculate this height in your text as in

"The average child's height is 5.84 inches."

The above line reads

The average child's height is **r mean(iris$Sepal.Length), 2)** inches.

where the bolded *r mean(iris$Sepal.Length), 2)* is placed into so that Markdown knows that it is an R command and that it should execute it. I bolded it here for the illustration purposes.

Additional comments

- Don't use the R variable names in your written part. If you make a histogram of via *hist(myData$hgb)*, you should not use *hgb* in the text. If, for example, *hgb* represents the hemoglobin blood levels, then you should use hemoglobin blood levels in your document.
- To interpret coefficients say something like

When the stove is turned off, our model predicts that 450.19 seconds are needed for 1200 g of snow to melt on average.

- Write down you estimated model:

The straight line relationship between the stove setting and the time needed to melt 120 g of snow is given by

$$temperature = 450.19 - 234 \times stove\ setting.$$

- Run the spell check! Each misspelled word is penalized. The spell check button is next to the knit button and it looks like ABC with a check below them.
- Round your values to something reasonable.
- Show your calculations, i.e. type them out. For example

$$\begin{aligned}
\Delta y &= y(x+1) - y(x) \\
&= \beta_0 + \beta_1(x+1) + \beta_2(x+1)^2 - \beta_0 + \beta_1 x + \beta_2 x^2 \\
&\vdots \\
&= \beta_1 + \beta_2 + 2\beta_2 x
\end{aligned}$$

- Write out your derivative in LaTeX. For example

$$\frac{\mathrm{d}}{\mathrm{d}x} y = -65e^{-x}$$

- To add space between two words in LaTeX's math environment use ~ as in

$$stovesetting$$

vs

$$stove\ setting$$

- Don't start your answer with "answer:" but do start them on a new line.
- Use LaTeX where you can.